CO3_AT3_MCQ(Bayesian and Computational Learning)

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* Indicates required question

MCQ

21. EM stands for: *

- ○ Error Minimization
- ◉ Expectation-Maximization
- ○ Estimated Model
- ○ Example Matching

22. The EM algorithm is particularly useful when: *

- ◉ Hidden or unobserved variables are present
- ○ No training data is available
- ○ All variables are known
- ○ Only binary data is available

19. In a Bayesian Belief Network, a node normally represents: *

- (selected) A random variable
- A training iteration
- A distance measure
- A program statement

18. A Bayesian Belief Network is represented using a: *

- (selected) Directed Acyclic Graph
- Stack
- Queue
- Binary Search Tree

14. Suppose the posterior probabilities of three hypotheses are H1=0.1, H2=0.7, and H3=0.2. Which hypothesis has the highest probability of being selected by the Gibbs algorithm? *

- H1
- (selected) H2
- H3
- All equally

17. Suppose P(Spam)=0.4 and P(Offer|Spam)=0.8. What is the value of P(Offer|Spam)P(Spam)? *

- ○ 0.16
- ○ 0.20
- ◉ 0.32
- ○ 1.20

1. Which of the following is the correct form of Bayes' theorem? *

- ◉ P(A|B) = P(A)P(B)
- ○ P(A|B) = P(B|A)P(A) / P(B)
- ○ P(A|B) = P(A) + P(B)
- ○ P(A|B) = P(A) - P(B)

10. Two models have the same prediction accuracy. Model A is much more complex than Model B. According to the MDL principle, which model is generally preferred? *

- ○ Model A
- ◉ Model B
- ○ Both are always rejected
- ○ Neither can be selected

16. Which of the following is a common application of the Naïve Bayes classifier? *

- (selected) Email spam classification
- Sorting numbers
- Image resizing
- File compression

11. The Bayes optimal classifier makes predictions by: *

- Selecting only one arbitrary hypothesis
- (selected) Considering hypotheses weighted by their posterior probabilities
- Using only prior probabilities
- Selecting the most complex hypothesis

8. A classifier observes 80 successful outcomes in 100 trials. What is the maximum likelihood estimate of the probability of success? *

- 0.08
- 0.20
- (selected) 0.80
- 1.25

25. During the Maximization (M) step, the algorithm: *

- (selected) Updates model parameters using the current estimates of hidden variables
- Removes training examples
- Selects a random hypothesis
- Calculates only classification accuracy

5. A hypothesis is said to be consistent with a training set when it: *

- Classifies all training examples correctly
- (selected) Classifies only positive examples correctly
- Classifies only negative examples correctly
- Has the maximum number of parameters

23. The two main steps of the EM algorithm are: *

- Selection and Mutation
- Encoding and Decoding
- (selected) Expectation and Maximization
- Training and Testing

4. The primary goal of concept learning is to: *

- (selected) Find a hypothesis that correctly classifies examples
- Increase the number of training examples
- Remove all negative examples
- Randomly generate hypotheses

12. Three hypotheses have posterior probabilities H1=0.2, H2=0.6, and H3=0.2. Which hypothesis has the greatest influence on the prediction? *

- [ ] H1
- [x] H2
- [ ] H3
- [ ] All equally

24. During the Expectation (E) step, the algorithm primarily: *

- [x] Estimates the expected values or probabilities of hidden variables
- [ ] Deletes hidden variables
- [ ] Randomly changes the dataset
- [ ] Stops the algorithm

15. The Naïve Bayes classifier assumes that features are: *

- [x] Conditionally independent given the class
- [ ] Completely dependent
- [ ] Always numerical
- [ ] Always equally important

20. Consider the network Rain → WetGrass. What does the arrow indicate? *

- ○ WetGrass causes Rain
- ◉ Rain directly influences WetGrass
- ○ Rain and WetGrass are always independent
- ○ Both variables have the same probability

6. Maximum Likelihood Estimation selects the parameter values that: *

- ○ Minimize the training set
- ◉ Maximize the likelihood of the observed data
- ○ Maximize model complexity
- ○ Minimize prior probability

13. The Gibbs algorithm selects a hypothesis according to its: *

- ○ Training-set size
- ◉ Posterior probability
- ○ Number of attributes
- ○ Classification error only

2. If P(A)=0.4, P(B|A)=0.5, and P(B)=0.2, what is P(A|B)? *

- ○ 0.2
- ○ 0.5
- ◉ 1.0
- ○ 0.8

9. The Minimum Description Length (MDL) principle prefers a model that: *

- ( ) Has maximum complexity
- (•) Minimizes the combined description length of the model and data
- ( ) Always has zero training error
- ( ) Uses the maximum number of features

3. In Bayesian learning, P(H) represents: *

- ( ) Posterior probability
- ( ) Likelihood
- (•) Prior probability
- ( ) Evidence

7. A coin is tossed 10 times and produces 7 heads. What is the maximum likelihood estimate of the probability of heads? *

- ( ) 0.3
- ( ) 0.5
- (•) 0.7
- ( ) 1.0

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